VIEW FROM THE GUN SIGHT

The second-largest military in the world, the Indian armed forces use assault rifles of various makes



INSAS (Indian Small Arms System): A gas-operated rifle that fires the NATO round—a 5.56 x 45 mm cartridge used as a standard rifle bullet for NATO forces.

- **400 metres**
- **1998**
- **Around a million**



AK-74: The AK-74 is a variant of the AK-47 that fires the 5.45x39mm high velocity round and is more accurate than the earlier AKM version.

- **500 metres**
- **Mid-2000s**
- **15,000**



M16A2: An American-made 5.56 x 45mm calibre assault rifle. Constructed from lightweight material, it has features like an adjustable rear sight

- **550 m**
- **Late 1980s**
- **1,000**

AK-47: Used in large numbers by the Indian Army, and widely used in counter-terrorist operations. Known for its sturdiness, fires a 7.62 x 39 mm bullet

- **350 m**
- **1990s**
- **250,000**



TAR-21 (Tavor): An Israeli-made assault rifle, it fires a 5.56 x 45 mm round. With its magazine behind the trigger, it has a shorter length, increasing manoeuvrability

- **500 m**
- **Early 2000s**
- **70,000**



- Range
- In use since
- Numbers in use

The key reason for the Indian Army's desperate effort to import more Sig Sauers was the undoing of the much-hyped India-Russia co-production of the AK-203 rifle due to reasons like costs and share of indigenous content. However, the INSAS isn't being replaced lock, stock and barrel. Due to the sheer numbers in use and the enormous cost involved in replacing them, efforts are under way for an upgraded version that seeks to rectify flaws and makes them serviceable for a longer period.

Speaking about the tortuous course of procuring/ manufacturing a dependable assault rifle, Lt Gen. D.S. Hooda (retd), former Northern Army commander, says that the DRDO was never really focused on the task at hand, while the Indian Army was never sure about its exact requirements. "Army headquarters was never clear on what type of weapon it was looking for. Manufacturers can come to you only when you decide what you want," he says. Furthermore, the army headquarters' complex General Staff Qualitative Requirements (GSQRs), which dictate the specifications and standards for new equipment, have confused Indian weapons manufacturers. The GSQRs for small arms have been criticised for their disconnect with the actual needs on the ground.

Lt Gen. Anil Ahuja (retd), former deputy chief of the Integrated Defence Staff for Policy Planning and Force Development, agrees, and says that India's inability to have an indigenous assault rifle for its armed forces is a cumulative result of the inability to clearly define operational requirements, the inability of the public sector rifle factory to manufacture a dependable weapon and slippages in the acquisition process.

However, defence officials maintain that any small arms weapon system takes a long time to evolve. The American M4 assault rifle, they point out as an instance, evolved over 50 years from its initial models, which were also marred by consistent quality issues.

ARC OF RIFLE FIRE

During the 1962 Sino-Indian war, Indian soldiers faced their AK-47-toting adversaries with outdated bolt-action Enfield .303 rifles. That blunderbuss was eventually replaced by the 7.62 x 51 mm SLR (self-loading rifle, an earlier avatar of the semi-automatic rifle), with which the Indian military fought the 1965 and 1971 wars against Pakistan and which served till the late '90s. In the '80s, in keeping with the global trend, a decision was taken to change the calibre of the standard service rifle from 7.62 mm to 5.56 mm. The reason was a paradigm shift in the doctrine of fighting—wounding the enemy on the battlefield, and making him a logistical liability for others, was deemed more advantageous than to take him out altogether. Shorter length and girth also make a bullet lighter, allowing for lesser penetration of the target upon contact and, therefore, a non-fatal injury. Moreover, the round produced less recoil, with a higher muzzle velocity and flatter trajectory, making it more accurate till about 200 metres—the common engagement range in a firefight. In addition, the bullet was lighter and cheaper to produce as it had less brass and lead. The new INSAS rifles, therefore, came with a 5.56 x 45 mm calibre.

However, as Lt Gen. Ahuja points out, "While the rationale behind switching from 7.62 to 5.56 calibre still holds good for conventional operations, it's inadequate for counter-terrorist opera-